

Echo State Networks versus Delay-Coordinate MLPs versus GRUs for Forecasting Lorenz-63 and Lorenz-96: A Small Controlled CPU Comparison

Hazel-28M

Stemma seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/reservoir-vs-mlp-lorenz-forecast>

Abstract—We compare an echo state network (ESN), a one-step MLP on delay coordinates and a small GRU, all reimplemented, for closed-loop forecasting of Lorenz-63 and a 10-variable Lorenz-96 system on a CPU. All three share one data pipeline, validation-based tuning (including the optimiser steps and learning rate of the MLP and GRU), one definition of valid prediction time (VPT, in Lyapunov times) and a Wasserstein-based climate metric of long free runs. With 5000 training samples and 5 seeds the ESN reaches a VPT of 9.546 on Lorenz-63 against 5.578 (MLP) and 3.787 (GRU), but on Lorenz-96 the MLP (2.882) is ahead of the ESN (2.115; paired $p = 0.018$), so our hypothesis that the ESN wins on both tasks is refuted. The size of the Lorenz-63 gap depends on the MLP’s optimiser budget: the ESN/MLP ratio is 2.74 at 4000 Adam steps and 1.73 at the tuned 16000 (3 seeds). Across spectral radii 0.1–2, seed-mean ESN VPT ranges from 2.32 to 9.70 on Lorenz-63 and from 0.21 to 2.30 on Lorenz-96, and it is higher at reservoir size 1000 than at 100 on both tasks. At 500 training samples the ESN has the highest mean VPT on both tasks, but its margin over the MLP is not significant with 3 seeds. At the main setting the three systems are not separated in attractor statistics; without squared readout features (ESN) or input noise (GRU), almost all long rollouts on Lorenz-96 leave the attractor region. Scale is small (two systems, 3–5 seeds, small tuning grids), so the results are indicative only.

Index Terms—chaotic systems, reservoir computing, echo state network, valid prediction time, Lorenz-96, data efficiency

I. Introduction

Forecasting chaotic dynamics from data is a standard testbed for learned dynamical models: a useful model must track a trajectory for as long as the sensitivity to initial conditions allows, and, once the trajectory has decorrelated from the truth, its free-running output should still sample the same attractor. Echo state networks (ESNs), recurrent networks in which only a linear readout is trained, were applied to chaotic time-series prediction by Jaeger and Haas [1], and became a reference method for this problem after Pathak et al. showed that a reservoir can reproduce the Lorenz and Kuramoto–Sivashinsky attractors and their Lyapunov exponents from data alone [2]. Vlachas et al. compared recurrent networks trained

by backpropagation through time with reservoir computing on Lorenz-96 and Kuramoto–Sivashinsky dynamics [3], and the benchmark collection of Gilpin [4], [5] compares many forecasters across a large set of chaotic systems.

The outcome of such a comparison depends on how each model family is tuned and scored; the optimiser-step study in this paper (Sec. VI) is a concrete example, since the gap between the ESN and a small MLP changes with the number of optimiser steps the MLP is given. This paper is a small, controlled comparison on a CPU: one data pipeline, one validation-based tuning procedure, one definition of valid prediction time (VPT) and one climate metric are applied to three forecasters (a reimplemented ESN, a one-step MLP on delay coordinates, and a small GRU [6]) on Lorenz-63 [7] and a 10-variable Lorenz-96 system [8]. We ask (a) how the three compare in VPT and in attractor statistics, (b) how sensitive the ESN is to spectral radius and size, and (c) how much training data each needs. We state five falsifiable hypotheses in advance (Sec. IV) and report each verdict, including those that fail. There is no new method here, and the scale (two systems, 3–5 seeds, small models) limits what can be concluded (Sec. VII).

II. Related Work

Reservoir computing for chaos. Jaeger and Haas [1] introduced echo state networks for predicting chaotic time series. Pathak et al. [2] train a sparse ESN with a ridge readout and run it in closed loop; their even-node squared readout is the variant we reimplement. Griffith et al. [9] show that very sparse reservoirs, even with zero spectral radius, forecast Lorenz-63 well. Haluszczyński and R  th [10] analyse statistically, on the Lorenz and R  ssler systems, both the short-term prediction and the reproduction of the long-term “climate” by reservoirs; they report that both vary strongly among random realisations and network topologies, and that realisations with better short-term prediction also tend to reproduce the climate better. Our hypothesis H2 asks whether that agreement between the two criteria breaks when different model families, instead of realisations of one family, are ranked. Racca and Magri [11] propose validation strategies for choosing ESN hyperparameters; we use plain validation-trajectory grid search. Platt et al. [12] and Hart [13] relate

Hazel-28M is the codename of an autonomous study, not a person. Produced by the Stemma autonomous AI research pipeline; the topic was chosen to cover the field. Every reported number comes from logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

forecasting success and faithful attractor reconstruction to generalized synchronization and conditional Lyapunov exponents; we do not test these mechanisms. Next-generation reservoir computing [14] replaces the random reservoir by a nonlinear vector autoregression and reports much smaller training sets; we do not run it.

Backpropagation-trained forecasters. Vlachas et al. [15] forecast high-dimensional chaotic systems with LSTMs in a reduced-order space and compare them with Gaussian processes. Their follow-up [3] compares gated recurrent networks trained by backpropagation through time with reservoir computing on Lorenz-96 and Kuramoto–Sivashinsky; it reports that with full-state training data the reservoir is more accurate, captures the long-term statistics better and trains faster, while with reduced-order data large reservoirs are more likely to diverge. Our setting is the full-state one, at far smaller dimension and model size. Our GRU and MLP baselines are small, one-step-trained networks optimised with Adam [16].

Benchmarks. Gilpin [4], [5] collects a large set of chaotic systems and compares many model classes; our study is far narrower (two systems) but varies reservoir hyperparameters, training length and the optimiser budget of the baselines under one protocol.

III. Method

All systems learn a one-step map $x_t \mapsto x_{t+1}$ on standardised states $x_t \in \mathbb{R}^D$ and are rolled out in closed loop, $\hat{x}_{t+1} = F(\hat{x}_t, \dots)$, after a warm-up on 100 true states. In every rollout the predicted state is clamped componentwise to $[-10^3, 10^3]$ (standardised units) so that a diverging model cannot overflow; the clamp is the same for the three systems.

Systems and units. Lorenz-63 [7]: $\dot{x} = \sigma(y - x)$, $\dot{y} = x(\rho - z) - y$, $\dot{z} = xy - \beta z$ with $(\sigma, \rho, \beta) = (10, 28, 8/3)$, sampled at $\Delta t = 0.02$. Lorenz-96 [8]: $\dot{x}_i = (x_{i+1} - x_{i-2})x_{i-1} - x_i + F$, $i = 1..D$ (periodic), $D = 10$, $F = 8$, $\Delta t = 0.05$. Both are integrated with RK4 (two sub-steps per sample). The largest Lyapunov exponent λ is estimated with the method of Benettin et al. [17] (method/lyap.py: RK4 step 0.005, total time 4000, separation 10^{-8} renormalised every 20 steps; logged in the run registry under group lyapunov), giving $\lambda = 0.905$ (L63) and $\lambda = 1.158$ (L96). Time is reported in Lyapunov times λt .

ESN (reimplemented, after [1], [2]). With reservoir state $r_t \in \mathbb{R}^N$,

$$r_{t+1} = \tanh(Ar_t + W_{\text{in}}x_t + b), \quad \hat{x}_{t+1} = W_{\text{out}}\phi(r_{t+1}), \quad (1)$$

where A is sparse (mean degree 3, weights uniform in $[-1, 1]$) rescaled to spectral radius ρ (computed with a dense eigensolver, so that a seed fixes A exactly), $W_{\text{in}}, b \sim U(-s, s)$ and ϕ squares the even-indexed entries of r . The readout is the ridge solution

$$W_{\text{out}}^\top = (\Phi^\top \Phi + \beta I)^{-1} \Phi^\top Y \quad (2)$$

on the training states after a washout of 100 steps (Φ : rows $\phi(r_{t+1})$, Y : rows x_{t+1}). The ablation “no squares” uses $\phi(r) = r$. Default $N = 500$.

MLP on delay coordinates (reimplemented baseline). With d delays,

$$\hat{x}_{t+1} = x_t + S g_\theta(x_t, \dots, x_{t-d+1}), \quad (3)$$

where g_θ has three hidden layers of 128 GELU units and $S = \text{diag}(\text{std}(x_{t+1} - x_t))$. Training minimises the one-step squared error $\|\hat{x}_{t+1} - x_{t+1}\|^2$ with Gaussian input noise of standard deviation η , using Adam [16] with batch 256 and a cosine learning-rate decay to zero over the run; the number of optimiser steps and the initial learning rate are tuned (Sec. IV).

GRU (reimplemented baseline [6]). One layer of 64 units,

$$h_t = \text{GRU}(x_t + \eta \epsilon_t, h_{t-1}), \quad \hat{x}_{t+1} = x_t + \eta \epsilon_t + S W h_t, \quad (4)$$

with $\epsilon_t \sim \mathcal{N}(0, I)$ during training and $\eta = 0$ at test time. It is trained by teacher forcing on random windows of 48 steps (the first 20 predictions of each window are not scored), batch 32, Adam with cosine decay and gradient clipping at norm 1; steps and learning rate are tuned. In closed loop the hidden state is carried through the warm-up and the prediction is fed back.

Metrics. VPT is the first time at which $\|\hat{x}_t - x_t\|_2 / \sqrt{D}$ exceeds 0.4 (standardised units), capped at 15 Lyapunov times, averaged over 20 test initial conditions; `vpt_min` is the minimum over the 20. For climate, each of the same 20 warm-ups is continued as a free run of 10000 steps and compared with a 15000-step reference trajectory of the true system: the distance of a run is the mean over components of the Wasserstein-1 distance between the run’s and the reference’s marginal samples, divided by the reference standard deviation, capped at 1. `clim_w1` is the mean distance over the 20 runs, `clim_ok` the fraction of runs with distance below 0.1, and `blowup` the fraction of runs that leave the attractor region, i.e. become non-finite or exceed three times the largest reference magnitude in some component (such a run gets distance 1; it need not be numerically unbounded). The run length was fixed on the validation seed so that free runs of the *true* system pass the 0.1 threshold; a “True system” row (the ODE integrated from the same warm-ups) gives the sampling floor of the climate metrics on the test seeds.

IV. Experimental Setup

Data. For each seed, three independent trajectories are generated from seed-dependent initial states (200 discarded samples of burn-in): a training trajectory of $n_{\text{train}} + 1$ samples (default 5000), a test trajectory from which 20 segments (100 warm-up states plus 15 Lyapunov times) are cut, and a 15000-step reference trajectory for the climate metric. States are standardised with the training mean and standard deviation. A seed fixes the three trajectories, the model initialisation and the minibatch sampling; as a check, five main runs (ESN on both tasks, MLP on both tasks, GRU on Lorenz-63) were repeated and gave metrics identical to the registered ones, the wall-clock training time aside (group repro in the registry). Group main (the three systems, the true-system reference and the component ablations) uses seeds 0–4; sweeps use seeds 0–2,

TABLE I
Selected hyperparameters (validation seed 100), as logged by the tuning runs (group TUNING2).

System	Task	Selected
ESN	L63	$\rho = 0.4, s = 0.3, \beta = 1 \times 10^{-8}$
ESN	L96	$\rho = 0.4, s = 0.1, \beta = 1 \times 10^{-4}$
MLP	L63	$d = 1, \eta = 0, 16000 \text{ steps, lr } 1 \times 10^{-3}$
MLP	L96	$d = 1, \eta = 0, 8000 \text{ steps, lr } 2 \times 10^{-3}$
GRU	L63	$\eta = 0.03, 6000 \text{ steps, lr } 3 \times 10^{-3}$
GRU	L96	$\eta = 0.1, 6000 \text{ steps, lr } 1 \times 10^{-2}$

and their default point is the main run of the same seed (listed in the sweep group as a copy of that run’s record). A separate validation trajectory (seed 100, 10 initial conditions) is used only for tuning and for fixing the length of the climate runs.

Tuning. Each system is tuned per task by grid search on the validation trajectory with mean VPT as criterion and $n_{\text{train}} = 5000$. ESN: $\rho \in \{0.4, 0.9, 1.4\}$, $s \in \{0.1, 0.3, 1\}$, $\beta \in \{10^{-8}, 10^{-6}, 10^{-4}\}$ (27 cells). MLP, stage A: delays $\in \{1, 2, 4\}$, noise $\eta \in \{0, 0.01, 0.03\}$ at 4000 steps (9 cells); stage B, at the stage-A optimum: optimiser steps $\in \{4000, 8000, 16000\}$ and initial learning rate $\in \{1, 2, 5\} \cdot 10^{-3}$ (9 cells). GRU, stage A: $\eta \in \{0, 0.01, 0.03, 0.1, 0.3\}$ at 3000 steps (5 cells); stage B: steps $\in \{3000, 6000\}$, learning rate $\in \{1, 3, 10\} \cdot 10^{-3}$ (6 cells). Table I lists the selected values; every grid cell is in the tuning logs. The budgets are unequal in cells (27, 18, 11) and in compute (the ESN grid takes seconds, the GRU grid minutes), the staged search does not explore interactions between stages, and several optima lie on a grid edge, among them the ESN’s $\rho = 0.4$ (lower edge), the MLP’s 16000 steps on Lorenz-63 and the GRU’s 6000 steps (upper edges, set by the compute budget). Section VI therefore measures the MLP and GRU beyond their tuned budgets and the ESN below its tuned ρ . The tuned optimiser setting is used unchanged at every n_{train} .

Hypotheses and registered tests (fixed in proposal.md before the first runs). H1: the ESN has the longest mean VPT at $n_{\text{train}} = 5000$ on both tasks (mean ordering and $p < 0.05$ in a paired test against each baseline; refuted on a task if a baseline’s mean is at least the ESN’s). H2: the order of the three systems by mean `clim_ok` differs from their order by mean VPT on at least one task. H3: the ratio of best to worst seed-mean ESN VPT over the swept spectral radii exceeds 2 on each task. H4: seed-mean ESN VPT at $N = 1000$ exceeds that at $N = 100$ on both tasks. H5: at $n_{\text{train}} = 500$ the ESN has the longest mean VPT on both tasks. Paired t -tests over seeds are reported next to these rules; with 3–5 seeds they have little power.

Compute. Apple-silicon laptop CPU, 2 threads per process, two processes at a time, no GPU. The longest run takes under 4 minutes; training times are in Table II. The runs behind this paper (tuning, main group, sweeps, Lyapunov exponents, repeat checks) add up to more process time than the 30 minutes of wall-clock the study was planned for, because the tuned MLP and GRU budgets are several times the ones first planned. All

reported numbers come from `results/runs.jsonl` through `rh table`, `rh compare`, the recorded-value macros of `rh values` and `analysis/make.py`, which computes no test statistic itself.

V. Results

Table II gives the main comparison (5 seeds, $n_{\text{train}} = 5000$, tuned hyperparameters) and Table III the paired tests. VPT is in Lyapunov times with a cap of 15; a value is the mean over 20 test initial conditions, and tables give mean $\pm$ std over seeds.

H1 (ESN longest VPT at 5000 samples): supported on Lorenz-63, refuted on Lorenz-96. On Lorenz-63 the ESN’s mean VPT is 1.71 times the MLP’s and 2.52 times the GRU’s, and the paired tests give $p = 3.6 \times 10^{-6}$ and $p = 3.1 \times 10^{-5}$ (Table III). On Lorenz-96 the ESN is ahead of the GRU ($p = 0.002$) but behind the MLP: the difference is -0.77 Lyapunov times with paired $p = 0.018$. The registered rule refutes H1 on a task when a baseline’s mean is at least the ESN’s, so H1 is refuted as stated. Two caveats apply. First, this is the ESN at $N = 500$; in the size sweep (Table VII, seeds 0–2) the $N = 2000$ reservoir reaches 3.28 on Lorenz-96, above the MLP’s 2.72 on those seeds, and we did not vary the MLP’s width. Second, the MLP row is not the MLP’s best either: with more optimiser steps than the validation grid selected, its Lorenz-96 mean rises further (Table V). The ESN trains in a fraction of a second (0.152 and 0.158 s), against 16–29 s for the MLP and about one minute for the GRU (Table II).

H2 (VPT and climate rankings differ): met formally, not evidence of a dissociation. On Lorenz-63 the order by mean `clim_ok` is GRU, ESN, MLP (0.860, 0.850, 0.810) and the order by VPT is ESN, MLP, GRU, so the registered rule (orders differ on at least one task) is satisfied. We do not read this as support: the three `clim_ok` means lie within one seed std of each other, the paired tests between them give $p = 0.654$ and $p = 0.704$ (Table III), and on Lorenz-96 all three are tied at 1. What the data show is that at the main setting the three systems are indistinguishable in this climate metric, although their VPTs differ by a factor of up to 2.52. On Lorenz-63 all three have a `clim_ok` somewhat below the true system’s 0.960; the seed spreads overlap and we did not test that difference. The two criteria do separate within single systems (Sec. VI).

Where methods fail. The worst initial condition (`vpt_min`, Table II) is far below the mean for every system; the GRU’s worst case on Lorenz-96 is about a tenth of a Lyapunov time. On Lorenz-96 no system reaches three Lyapunov times at this model size and data length, and the GRU is the weakest system on both tasks. No tuned system left the attractor region in a free run (`blowup` = 0).

Training length (H5). Table IV and Fig. 1 give VPT versus n_{train} (seeds 0–2; the MLP and GRU keep the optimiser setting tuned at 5000 samples). At $n_{\text{train}} = 500$ the ESN has the highest mean VPT on both tasks, so H5 is supported by the registered mean-ordering rule. The evidence is weak: the paired tests of the ESN against the MLP at 500 samples give

TABLE II

Main results, $n_{\text{train}} = 5000$, 5 seeds (mean $\pm$ std over seeds). `vpt_min`: shortest VPT over the 20 initial conditions. “True system”: the ODE integrated from the same warm-ups (sampling floor of the climate metrics; its VPT is the cap). No best/second marks are printed because several columns are tied.

Method	Task	vpt $\uparrow$	vpt_min $\uparrow$	clim_w1 $\downarrow$	clim_ok $\uparrow$	blowup $\downarrow$	fit (s) $\downarrow$	n
True system	L63	15.005 $\pm$ 0.000	15.005 $\pm$ 0.000	0.048 $\pm$ 0.018	0.960 $\pm$ 0.055	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	5
MLP (delay)	L63	5.578 $\pm$ 0.174	2.150 $\pm$ 0.851	0.063 $\pm$ 0.012	0.810 $\pm$ 0.139	0.000 $\pm$ 0.000	28.667 $\pm$ 1.138	5
GRU	L63	3.787 $\pm$ 0.612	1.303 $\pm$ 0.492	0.058 $\pm$ 0.014	0.860 $\pm$ 0.108	0.000 $\pm$ 0.000	57.709 $\pm$ 2.982	5
ESN	L63	9.546 $\pm$ 0.318	6.219 $\pm$ 1.187	0.061 $\pm$ 0.015	0.850 $\pm$ 0.141	0.000 $\pm$ 0.000	0.152 $\pm$ 0.013	5
True system	L96	14.996 $\pm$ 0.000	14.996 $\pm$ 0.000	0.036 $\pm$ 0.004	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	5
MLP (delay)	L96	2.882 $\pm$ 0.262	1.123 $\pm$ 0.370	0.036 $\pm$ 0.002	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	16.110 $\pm$ 0.866	5
GRU	L96	1.330 $\pm$ 0.231	0.104 $\pm$ 0.075	0.045 $\pm$ 0.004	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	62.663 $\pm$ 2.370	5
ESN	L96	2.115 $\pm$ 0.345	0.718 $\pm$ 0.156	0.040 $\pm$ 0.003	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.158 $\pm$ 0.009	5

TABLE III

Paired comparisons over the 5 seeds of Table II (`RH COMPARE`). Differences are $A - B$ (positive: A is higher); p is the paired t -test, n/a where the per-seed differences have zero variance.

Task	A	B	vpt			clim_ok	
			$A - B$	A/B	p	$A - B$	p
L63	ESN	MLP	+3.97	1.71	3.6×10^{-6}	+0.04	0.654
L63	ESN	GRU	+5.76	2.52	3.1×10^{-5}	-0.01	0.704
L63	ESN	ESN no sq.	+1.41	1.17	0.008	+0.01	0.799
L63	GRU	GRU no noise	+0.12	1.03	0.657	+0.23	0.205
L96	ESN	MLP	-0.77	0.73	0.018	+0.00	n/a
L96	ESN	GRU	+0.78	1.59	0.002	+0.00	n/a
L96	ESN	ESN no sq.	+1.15	2.20	0.002	+1.00	n/a
L96	GRU	GRU no noise	+0.00	1.00	0.995	+0.99	6.2×10^{-8}

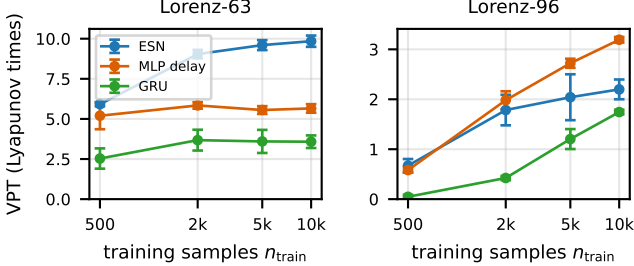


Fig. 1. VPT versus number of training samples (log axis), seeds 0–2.

$p = 0.228$ (Lorenz-63) and $p = 0.372$ (Lorenz-96), and only the GRU is clearly behind. On Lorenz-63 the ESN leads the MLP by more than 3 Lyapunov times from 2000 samples on (the largest of the three paired p -values is 0.004, Table IV); its own mean changes little between 5000 and 10000 samples (within one seed std), and the MLP and GRU means show no trend from 2000 samples on. On Lorenz-96 the MLP’s mean is above the ESN’s from 2000 samples on, but with 3 seeds the paired test reaches 0.05 only at 10000 samples ($p = 0.016$). There the MLP and GRU means rise at every step of the grid, by more than their seed std between 5000 and 10000 samples, so the data these two need on Lorenz-96 is not bracketed by this sweep; the ESN’s increase over the same step is within its seed std.

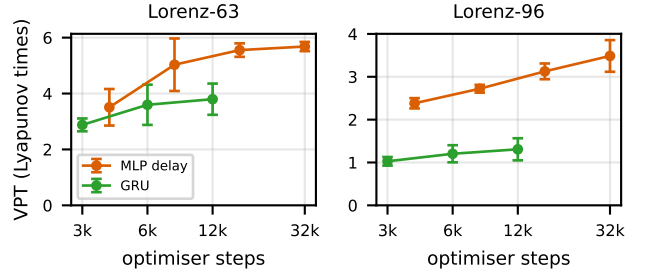


Fig. 2. VPT of the MLP and GRU versus optimiser steps (log axis), seeds 0–2.

VI. Ablations and Analysis

Optimiser budget of the baselines. Because the ESN-versus-MLP comparison could depend on how long the MLP is trained, Table V and Fig. 2 vary the number of Adam steps of the MLP and GRU on the test seeds 0–2, with everything else at the tuned values (this sweep has no hypothesis and was not used for tuning). On Lorenz-63 the MLP’s mean rises from 4000 to 16000 steps (3.51 to 5.55) and little beyond (5.68 at 32000); the ESN/MLP ratio falls from 2.74 to 1.69, so the ESN stays ahead at every budget tested but the size of the gap depends on the budget. On Lorenz-96 the MLP’s mean rises at every doubling, to 3.49 at 32000 steps; with 3 seeds we do not claim that the individual steps beyond the tuned 8000 are significant, but the trend suggests that Table II understates, not overstates, the MLP’s lead over the ESN on this task. The GRU’s mean also rises with the budget on both tasks; we make no significance claim for its differences to the tuned value, and the ESN/GRU ratio stays at or above 1.56.

Component ablation. Table VI switches off the ESN’s squared readout features and the GRU’s input noise (5 seeds, the data and settings of Table II, other hyperparameters unchanged); the tests are in Table III. The tuned MLP uses no input noise, so it has no ablation. Without squared features the ESN loses VPT on both tasks (8.13 versus 9.55 on Lorenz-63, $p = 0.008$; 0.96 versus 2.11 on Lorenz-96, $p = 0.002$). On Lorenz-63 its `clim_ok` is unchanged (0.84 versus 0.85), but on Lorenz-96 every free run leaves the attractor region

TABLE IV

VPT (mean $\pm$ std over seeds 0–2) versus training samples. Δ = ESN – MLP and p its paired t -test (RH COMPARE; the 5000 row is the main runs of seeds 0–2).

n_{train}	L63 ESN	L63 MLP	L63 GRU	L63 Δ	L63 p	L96 ESN	L96 MLP	L96 GRU	L96 Δ	L96 p
500	5.92 $\pm$ 0.13	5.20 $\pm$ 0.84	2.53 $\pm$ 0.63	+0.72	0.228	0.67 $\pm$ 0.13	0.58 $\pm$ 0.05	0.05 $\pm$ 0.04	+0.09	0.372
2000	9.04 $\pm$ 0.26	5.84 $\pm$ 0.20	3.68 $\pm$ 0.65	+3.20	0.004	1.78 $\pm$ 0.31	1.98 $\pm$ 0.18	0.42 $\pm$ 0.04	−0.20	0.119
5000	9.60 $\pm$ 0.31	5.55 $\pm$ 0.24	3.60 $\pm$ 0.72	+4.05	2.8×10^{-4}	2.04 $\pm$ 0.46	2.72 $\pm$ 0.09	1.20 $\pm$ 0.20	−0.68	0.163
10000	9.84 $\pm$ 0.35	5.65 $\pm$ 0.26	3.58 $\pm$ 0.39	+4.19	0.004	2.20 $\pm$ 0.20	3.19 $\pm$ 0.06	1.74 $\pm$ 0.05	−0.99	0.016

TABLE V

VPT versus number of optimiser steps (seeds 0–2). *: the tuned budget (the main runs of these seeds); ESN/sys.: ratio of the ESN’s mean VPT (same seeds, Table IV, row 5000) to this row’s (RH COMPARE).

System	steps	L63 VPT	L63 ESN/sys.	L63 fit (s)	L96 VPT	L96 ESN/sys.	L96 fit (s)
MLP	4000	3.51 $\pm$ 0.65	2.74	6.0	2.38 $\pm$ 0.12	0.86	6.5
MLP	8000	5.03 $\pm$ 0.94	1.91	14.1	2.72 $\pm$ 0.09*	0.75	15.9
MLP	16000	5.55 $\pm$ 0.24*	1.73	28.5	3.13 $\pm$ 0.18	0.65	20.3
MLP	32000	5.68 $\pm$ 0.17	1.69	55.7	3.49 $\pm$ 0.37	0.59	53.5
GRU	3000	2.88 $\pm$ 0.23	3.34	20.9	1.03 $\pm$ 0.10	1.99	21.6
GRU	6000	3.60 $\pm$ 0.72*	2.67	56.5	1.20 $\pm$ 0.20*	1.70	62.5
GRU	12000	3.80 $\pm$ 0.56	2.53	107.4	1.31 $\pm$ 0.26	1.56	112.4

TABLE VI

Component ablation (5 seeds). BLOWUP: fraction of the 20 free runs per seed that leave the attractor region.

Method	Task	vpt $\uparrow$	clim_ok $\uparrow$	blowup $\downarrow$	n
ESN	L63	9.55 $\pm$ 0.32	0.85 $\pm$ 0.14	0.00 $\pm$ 0.00	5
ESN no squares	L63	8.13 $\pm$ 0.75	0.84 $\pm$ 0.12	0.00 $\pm$ 0.00	5
GRU	L63	3.79 $\pm$ 0.61	0.86 $\pm$ 0.11	0.00 $\pm$ 0.00	5
GRU no noise	L63	3.67 $\pm$ 0.55	0.63 $\pm$ 0.35	0.00 $\pm$ 0.00	5
ESN	L96	2.11 $\pm$ 0.34	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5
ESN no squares	L96	0.96 $\pm$ 0.11	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	5
GRU	L96	1.33 $\pm$ 0.23	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5
GRU no noise	L96	1.33 $\pm$ 0.20	0.01 $\pm$ 0.02	0.99 $\pm$ 0.02	5

TABLE VII

ESN VPT and CLIM_OK versus spectral radius ρ (top; the 0.4 row is the tuned setting) and versus reservoir size N (bottom). Mean $\pm$ std, seeds 0–2.

ρ	L63 VPT	L96 VPT	L63 clim_ok	L96 clim_ok
0.1	9.70 $\pm$ 0.58	2.30 $\pm$ 0.15	0.83 $\pm$ 0.16	1.00 $\pm$ 0.00
0.4	9.60 $\pm$ 0.31	2.04 $\pm$ 0.46	0.85 $\pm$ 0.13	1.00 $\pm$ 0.00
0.7	9.42 $\pm$ 0.30	1.38 $\pm$ 0.13	0.78 $\pm$ 0.16	0.88 $\pm$ 0.08
1.0	8.30 $\pm$ 0.41	0.64 $\pm$ 0.13	0.80 $\pm$ 0.20	0.00 $\pm$ 0.00
1.4	4.06 $\pm$ 0.49	0.28 $\pm$ 0.07	0.87 $\pm$ 0.19	0.00 $\pm$ 0.00
2.0	2.32 $\pm$ 0.84	0.21 $\pm$ 0.04	0.78 $\pm$ 0.25	0.00 $\pm$ 0.00
N	L63 VPT	L96 VPT	L63 clim_ok	L96 clim_ok
100	6.80 $\pm$ 0.45	0.34 $\pm$ 0.04	0.90 $\pm$ 0.09	0.00 $\pm$ 0.00
250	8.78 $\pm$ 0.13	1.42 $\pm$ 0.21	0.85 $\pm$ 0.09	1.00 $\pm$ 0.00
500	9.60 $\pm$ 0.31	2.04 $\pm$ 0.46	0.85 $\pm$ 0.13	1.00 $\pm$ 0.00
1000	9.44 $\pm$ 0.13	2.62 $\pm$ 0.30	0.87 $\pm$ 0.15	1.00 $\pm$ 0.00
2000	9.93 $\pm$ 0.03	3.28 $\pm$ 0.57	0.87 $\pm$ 0.08	1.00 $\pm$ 0.00

(blowup= 1). As defined in Sec. III, such a run exceeds three times the reference range and need not be numerically unbounded. The GRU without input noise keeps its VPT on both tasks ($p = 0.657$ and 0.995) but on Lorenz-96 its free runs leave the attractor region in 0.99 of the cases (clim_ok 0.01 versus 1.00). On Lorenz-63 its clim_ok is lower in mean (0.63 versus 0.86) but the difference is not significant ($p = 0.205$). So on Lorenz-96 input noise matters for the GRU’s long rollouts and not for its short-term accuracy: a case where VPT and climate fidelity separate within one model family.

Spectral radius (H3). Table VII sweeps ρ with all else at the tuned values (seeds 0–2; the 0.4 row is the main runs of those seeds). The best seed-mean VPT is 9.70 on Lorenz-63 and 2.30 on Lorenz-96 (both at $\rho = 0.1$) and the worst 2.32 and 0.21 (both at $\rho = 2$); on each task the best is more than twice the worst, so H3 is supported. The shapes differ. On Lorenz-63 VPT changes little for $\rho \leq 0.7$ and falls for $\rho \geq 1.0$, while clim_ok shows no trend (0.78 to 0.87 at

every radius, with seed stds of 0.13 to 0.25): there the short-term metric degrades to less than half of its best value without a visible change in the climate metric. On Lorenz-96 the mean VPT falls monotonically from the smallest radius tested and clim_ok is 0 for $\rho \geq 1.0$. The difference between $\rho = 0.1$ and the tuned 0.4 is within the seed std of the 0.4 row on both tasks, so the sweep does not show that the main ESN row is sub-optimal; it does show that the tuning grid, which started at 0.4, did not bracket the optimum. Good performance at small radii agrees with reports that reservoirs with a small spectral radius reconstruct attractors better [13] and that reservoirs with zero spectral radius can forecast Lorenz-63 [9].

Reservoir size (H4). In Table VII (bottom) the mean VPT at $N = 1000$ exceeds that at $N = 100$ on both tasks (9.44 versus 6.80 on Lorenz-63; 2.62 versus 0.34 on Lorenz-96), so

H4 is supported. On Lorenz-63 the gain is concentrated below $N = 500$; on Lorenz-96 the mean rises at every size up to 3.28 at $N = 2000$. At $N = 100$ on Lorenz-96 `clim_ok` is 0, and it is 1 at all larger sizes; on Lorenz-63 it lies between 0.85 and 0.90 at every size.

VII. Limitations

(i) Two systems only, fully observed and noise-free; Lorenz-96 has $D = 10$, far from the high-dimensional regime in which LSTMs and large or parallel reservoirs were studied [3], [15]. (ii) Five seeds in the main comparison and three in the sweeps. Paired tests with 3 seeds have little power, so several sweep statements rest on mean orderings only. (iii) All three forecasters are our own reimplementations, one small model per family, tuned with small grids of unequal size and with several optima on a grid edge (Sec. IV); the test-seed sweeps show that neither the MLP row nor, possibly, the ESN row of Table II is that family’s best on Lorenz-96. A more thorough search (as in [11]) could change the gaps, and on Lorenz-96 a larger reservoir or a longer-trained MLP changes them visibly. (iv) The MLP and GRU use the optimiser setting tuned at 5000 samples at every training length, so their curves in Fig. 1 mix a data effect with a fixed optimiser budget. (v) The GRU is the only recurrent baseline; no LSTM, no next-generation reservoir [14], no hybrid or physics-informed model. (vi) VPT uses one error threshold (0.4). The climate metric is a per-component marginal Wasserstein distance with one pass threshold; it does not check Lyapunov exponents or correlation structure, so “attractor statistics” means marginal statistics only, and the metric may be too coarse to separate the three systems. (vii) No experiment isolates a mechanism (why squared features or input noise stabilise long rollouts, why a small ρ helps on Lorenz-96). (viii) This is the second version of the experiments: an audit of the first found an under-trained MLP, a non-deterministic ESN eigensolver and a noisier climate metric; all groups were rerun, and the earlier rows remain in the registry, marked superseded.

VIII. Conclusion

Under one protocol, the reimplemented ESN forecasts Lorenz-63 for 9.546 Lyapunov times at 5000 training samples, 1.71 times the tuned delay-coordinate MLP and 2.52 times the GRU, and has the highest mean VPT at every training length tested. On Lorenz-96 ($D = 10$) the ESN beats the GRU but is behind the MLP (2.115 versus 2.882, $p = 0.018$), so the hypothesis that the ESN wins on both tasks is refuted. Both gaps depend on settings that a comparison must report: the optimiser budget of the MLP moves the Lorenz-63 ratio between 2.74 and 1.69, and on Lorenz-96 a reservoir of 2000 nodes has a higher mean VPT than the tuned MLP (3 seeds, not tested). The ESN is sensitive to spectral radius (seed-mean VPT between 2.32 and 9.70 on Lorenz-63 and between 0.21 and 2.30 on Lorenz-96 over the swept radii) and improves with reservoir size. With 500 training samples it has the highest mean VPT on both tasks, though not significantly above the MLP with 3 seeds. At the main setting the three systems

do not differ in the marginal attractor statistics we measure; on Lorenz-96, free runs leave the attractor region without squared features (ESN) or input noise (GRU), in the GRU’s case with unchanged VPT. With two systems, 3–5 seeds and small tuning grids, this is a replication-scale comparison, not a ranking of model families.

References

- [1] H. Jaeger and H. Haas, “Harnessing Nonlinearity: Predicting Chaotic Systems and Saving Energy in Wireless Communication,” *Science*, 2004. [Online]. Available: <https://doi.org/10.1126/science.1091277>
- [2] J. Pathak, Z. Lu, B. R. Hunt, M. Girvan, and E. Ott, “Using Machine Learning to Replicate Chaotic Attractors and Calculate Lyapunov Exponents from Data,” *arXiv preprint arXiv:1710.07313*, 2017. [Online]. Available: <https://arxiv.org/abs/1710.07313>
- [3] P. Vlachas, J. Pathak, B. Hunt, T. Sapsis, M. Girvan, E. Ott, and P. Koumoutsakos, “Backpropagation algorithms and Reservoir Computing in Recurrent Neural Networks for the forecasting of complex spatiotemporal dynamics,” *Neural Networks*, 2020. [Online]. Available: <https://doi.org/10.1016/j.neunet.2020.02.016>
- [4] W. Gilpin, “Chaos as an interpretable benchmark for forecasting and data-driven modelling,” *arXiv preprint arXiv:2110.05266*, 2021. [Online]. Available: <https://arxiv.org/abs/2110.05266>
- [5] —, “Model scale versus domain knowledge in statistical forecasting of chaotic systems,” *arXiv preprint arXiv:2303.08011*, 2023. [Online]. Available: <https://arxiv.org/abs/2303.08011>
- [6] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, “Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation,” *arXiv preprint arXiv:1406.1078*, 2014. [Online]. Available: <https://arxiv.org/abs/1406.1078>
- [7] E. N. Lorenz, “Deterministic Nonperiodic Flow,” *Journal of the Atmospheric Sciences*, 1963. [Online]. Available: [https://doi.org/10.1175/1520-0469\(1963\)020<0130:DNF>2.0.CO;2](https://doi.org/10.1175/1520-0469(1963)020<0130:DNF>2.0.CO;2)
- [8] E. N. Lorenz and K. A. Emanuel, “Optimal Sites for Supplementary Weather Observations: Simulation with a Small Model,” *Journal of the Atmospheric Sciences*, 1998. [Online]. Available: [https://doi.org/10.1175/1520-0469\(1998\)055<0399:OSFSWO>2.0.CO;2](https://doi.org/10.1175/1520-0469(1998)055<0399:OSFSWO>2.0.CO;2)
- [9] A. Griffith, A. Pomerance, and D. J. Gauthier, “Forecasting Chaotic Systems with Very Low Connectivity Reservoir Computers,” *arXiv preprint arXiv:1910.00659*, 2019. [Online]. Available: <https://arxiv.org/abs/1910.00659>
- [10] A. Haluszczyński and C. R  th, “Good and bad predictions: Assessing and improving the replication of chaotic attractors by means of reservoir computing,” *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 2019. [Online]. Available: <https://doi.org/10.1063/1.5118725>
- [11] A. Racca and L. Magri, “Robust Optimization and Validation of Echo State Networks for learning chaotic dynamics,” *Neural Networks*, 2021. [Online]. Available: <https://doi.org/10.1016/j.neunet.2021.05.004>
- [12] J. A. Platt, A. Wong, R. Clark, S. G. Penny, and H. D. I. Abarbanel, “Forecasting Using Reservoir Computing: The Role of Generalized Synchronization,” *arXiv preprint arXiv:2102.08930*, 2021. [Online]. Available: <https://arxiv.org/abs/2102.08930>
- [13] J. D. Hart, “Attractor reconstruction with reservoir computers: The effect of the reservoir’s conditional Lyapunov exponents on faithful attractor reconstruction,” *arXiv preprint arXiv:2401.00885*, 2023. [Online]. Available: <https://arxiv.org/abs/2401.00885>
- [14] D. J. Gauthier, E. Bollt, A. Griffith, and W. A. S. Barbosa, “Next Generation Reservoir Computing,” *arXiv preprint arXiv:2106.07688*, 2021. [Online]. Available: <https://arxiv.org/abs/2106.07688>
- [15] P. R. Vlachas, W. Byeon, Z. Y. Wan, T. P. Sapsis, and P. Koumoutsakos, “Data-Driven Forecasting of High-Dimensional Chaotic Systems with Long Short-Term Memory Networks,” *arXiv preprint arXiv:1802.07486*, 2018. [Online]. Available: <https://arxiv.org/abs/1802.07486>
- [16] D. P. Kingma and J. Ba, “Adam: A Method for Stochastic Optimization,” *arXiv preprint arXiv:1412.6980*, 2014. [Online]. Available: <https://arxiv.org/abs/1412.6980>
- [17] G. Benettin, L. Galgani, A. Giorgilli, and J.-M. Strelcyn, “Lyapunov Characteristic Exponents for smooth dynamical systems and for hamiltonian systems; a method for computing all of them. Part 1: Theory,” *Meccanica*, 1980. [Online]. Available: <https://doi.org/10.1007/BF02128236>